Proposal: *Impact of health insurance and household income on teenage HPV immunization rates*

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Morignot, A.-E. and Kipp, O.
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Human papillomavirus infections can result in cancers of the cervix, vagina, vulva, penis, anus and back of the throat [1]. According to the Centers for Disease Control and Prevention (CDC), over 90% of cancers caused by HPV can be prevented by vaccination [1]. We hypothesize that teen HPV vaccination rates are positively correlated with median household income and health insurance coverage in Colorado counties.

In order to probe the potential relationship between teen HPV vaccination rates in Colorado counties and the proportion of teens that are uninsured as well as the associated median income, we plan to utilize data sourced from *Exploring Cancer in Colorado (ECCO)*, an interactive platform created by the University of Colorado Anschutz Cancer Center [2]. We will be utilizing county-level data. Additionally, we will be using population census data sourced from the United States Census Bureau [3, 4]. We will specifically select data from age ranges 9-17 to represent the age range at which it is recommended to receive HPV vaccination [5]. The teen HPV vaccination rate data was collected in 2023, therefore we will be extracting the United States Census Bureau data from the corresponding year. We will not be separating the household income data by type of household.

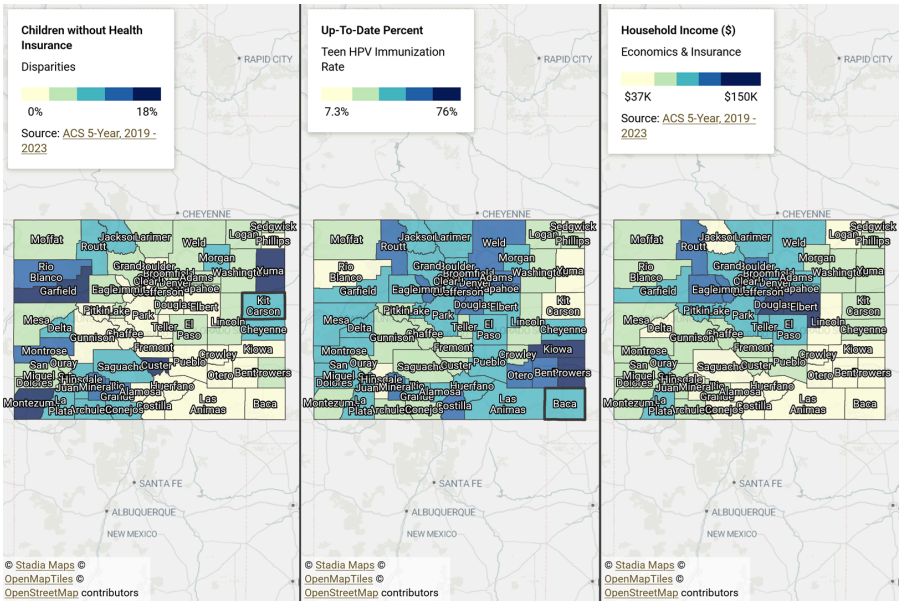

Figure 1. *Preliminary Representation of Project Proposal Data.* The above figure contains several county-level maps of the Colorado area, color-coded to represent differences in health insurance access, HPV vaccination status, and household income. Graphic was generated using ECCO [1].

The group members for this project consist of Ange Morignot and Oliver Kipp. The group would like to extend a special thank you to Marcus Richards for recommending *Exploring Cancer in Colorado* [1].

References

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3. U.S. Census Bureau, U.S. Department of Commerce. (n.d.). Population Under 18 Years by Age. *American Community Survey, ACS 5-Year Estimates Detailed Tables, Table B09001*. Retrieved February 9, 2026, from https://data.census.gov/table/ACSDT5Y2023.B09001?q=B09001:+Population+Under+18+Years+by+Age&g=040XX00US08,08$0500000.
4. U.S. Census Bureau. (n.d.). Income in the Past 12 Months (in 2024 Inflation-Adjusted Dollars). *American Community Survey, ACS 1-Year Estimates Subject Tables, Table S1901*. Retrieved February 9, 2026, from https://data.census.gov/table/ACSST1Y2024.S1901?g=040XX00US08,08$0500000.
5. Centers for Disease Control and Prevention. (n.d.). *Human Papillomavirus (HPV) Vaccination: Information for Health Care Providers*. Centers for Disease Control and Prevention. Retrieved February 9, 2026, from https://www.cdc.gov/vaccines/hcp/by-disease/hpv.html?CDC_AA_refVal=https%3A%2F%2Fwww.cdc.gov%2Fvaccines%2Fvpd%2Fhpv%2Fhcp%2Frecommendations.html.